

NPTEL Course on "Computer System Optimization for ML"

Lecture #31

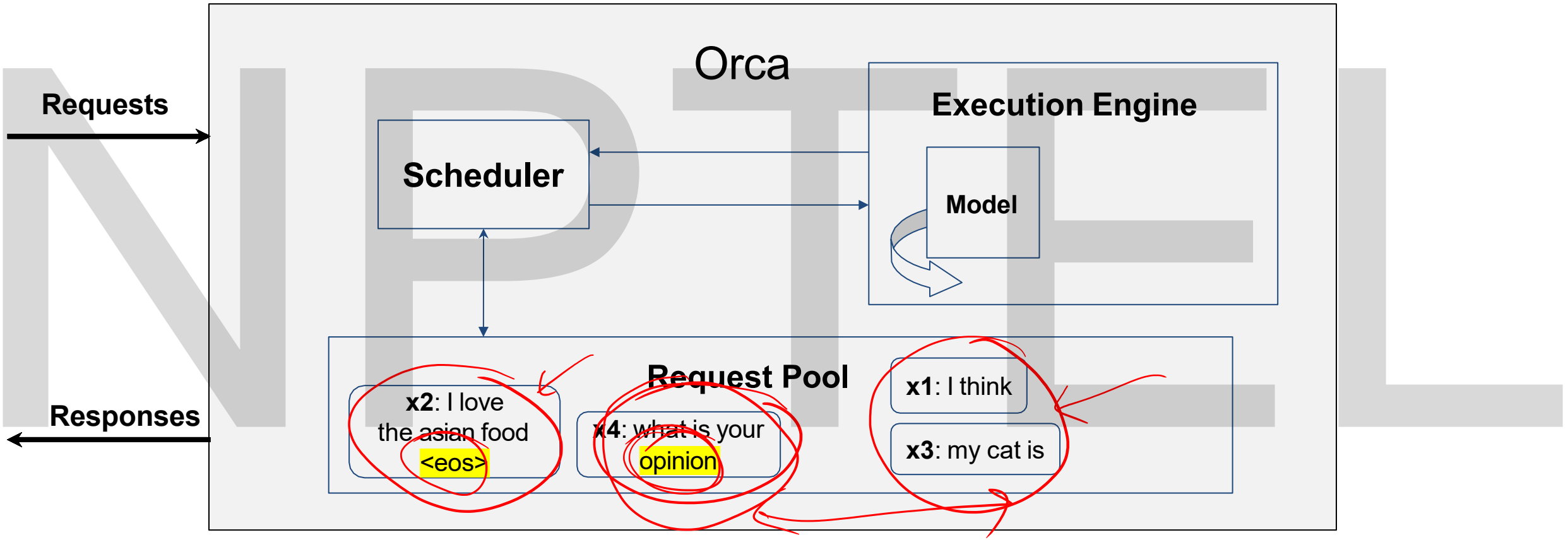
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Agenda

- Orca (contd.)
- Preliminary to batching

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Timestep 5



* Assume we set the max_batch_size = 3

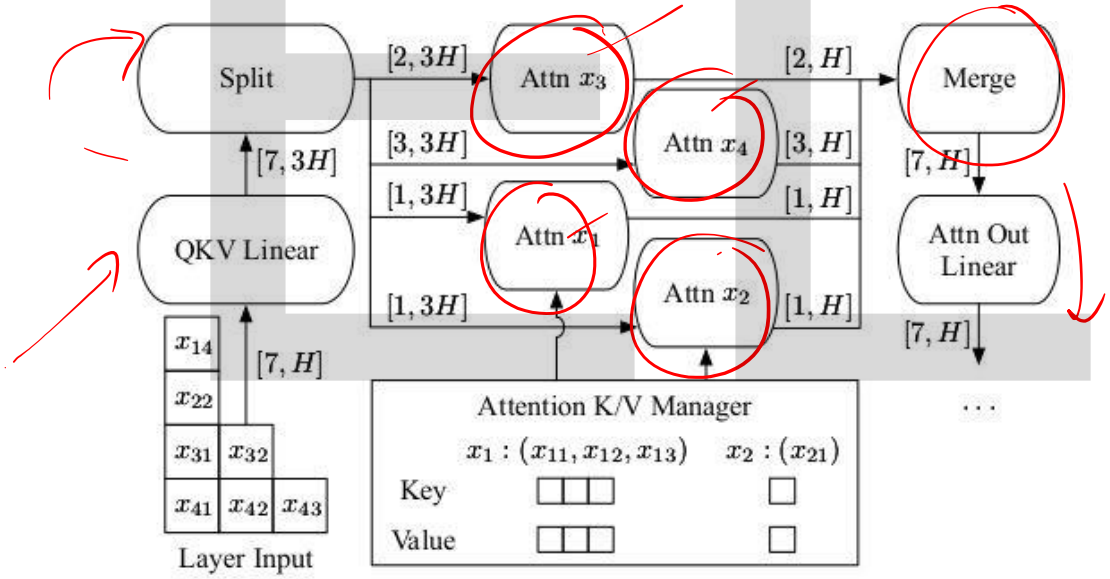
Solution: Selective Batching

Instead of batching **all** tensor operations, only batch the ones you can.

Flatten input tensors into 2-D tensor of shape [∑L, H], and feed into non-Attention operations (Linear, LayerNorm, Add, GeLU).

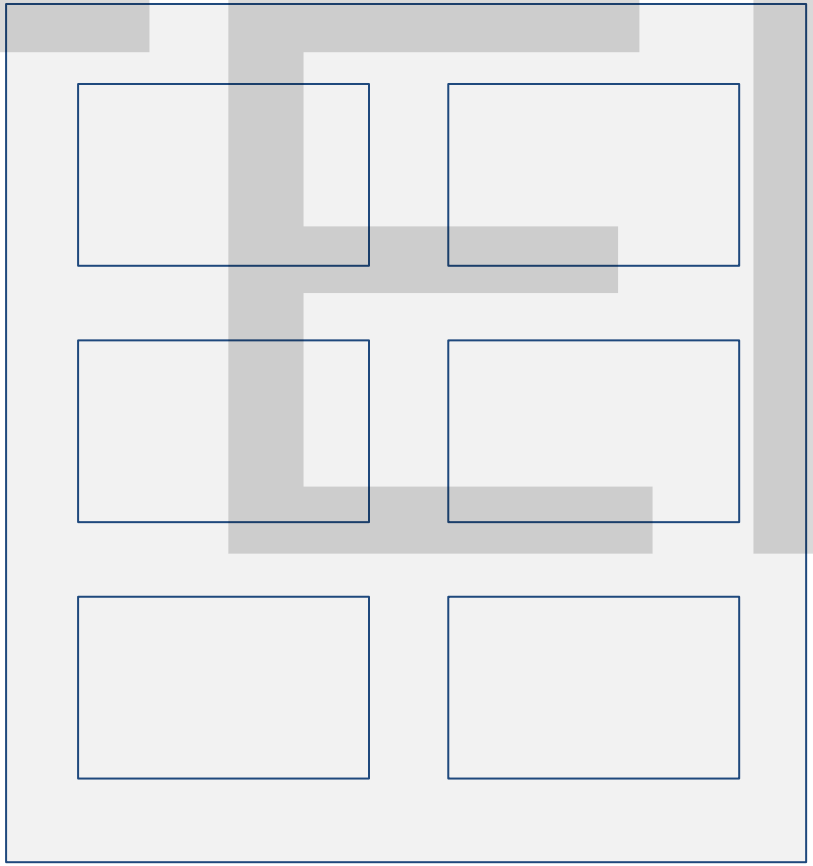
For Attention operations, split flattened tensor into individual requests and run Attention operation separately. Once completed, merge requests back.

Attention K/V manager provides Keys/Values separately for each request.



Scheduling Algorithm

Attention K/V Manager



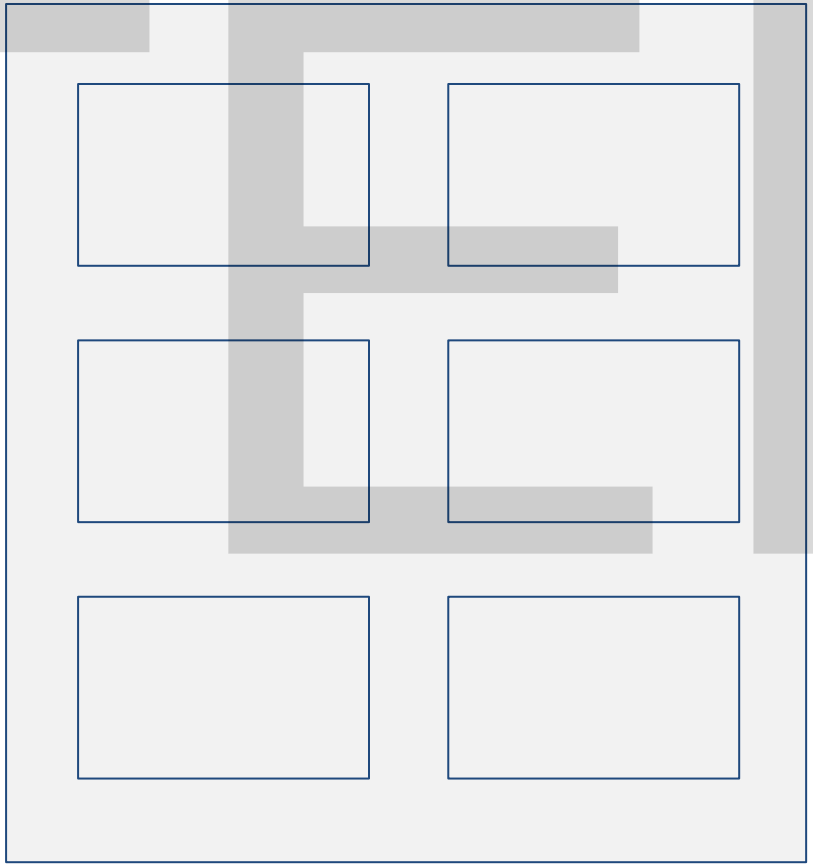
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Scheduling Algorithm

Iteration 1 …

x1: my cat is

Attention K/V Manager



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Scheduling Algorithm

Iteration 1 …

x1: my cat is

Attention K/V Manager



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Scheduling Algorithm

Iteration 1 …

x1: my cat is

Iteration 2 …

x1: I

Attention K/V Manager



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Scheduling Algorithm

Iteration 1 …



Iteration 2 …



Attention K/V Manager



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Scheduling Algorithm

Iteration 1

x1: my cat is so

Iteration 2 …

x1: I

Attention K/V Manager



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Scheduling Algorithm

Iteration 1

x1: my cat is so

Iteration 2

x1: I like

Attention K/V Manager



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Scheduling Algorithm

Iteration 1

x1: my cat is so

Iteration 2

x1: I like

Iteration 3

Deadlock!!

Attention K/V Manager



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Scheduling Algorithm

Solution: **Memory Aware Allocation** - reserve enough GPU space for all iterations at the 1st iteration itself

Predetermine ‘max_tokens’ any request can generate

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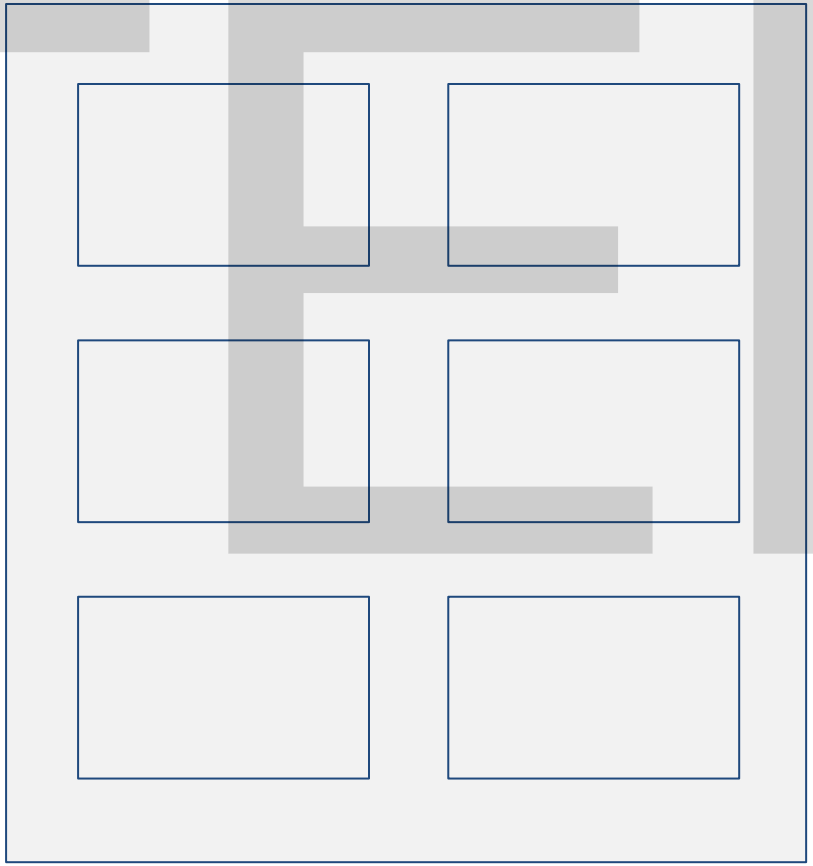
Scheduling Algorithm

Iteration 1 …

x1: my cat is

eg: max_tokens = 5

Attention K/V Manager



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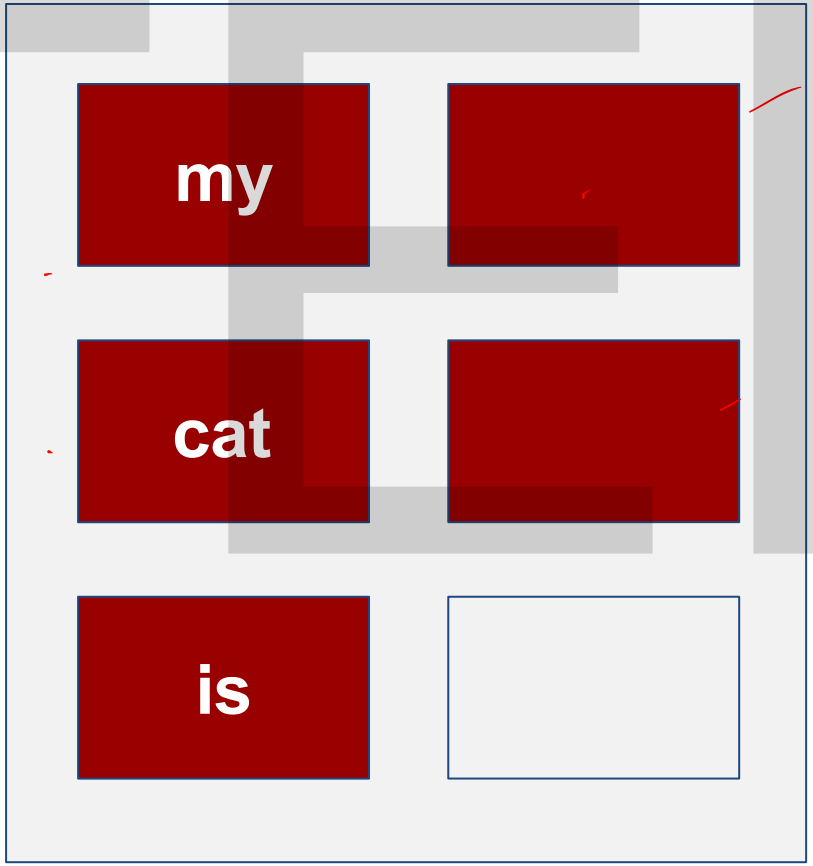
Scheduling Algorithm

Iteration 1 …

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eg: max_tokens = 5

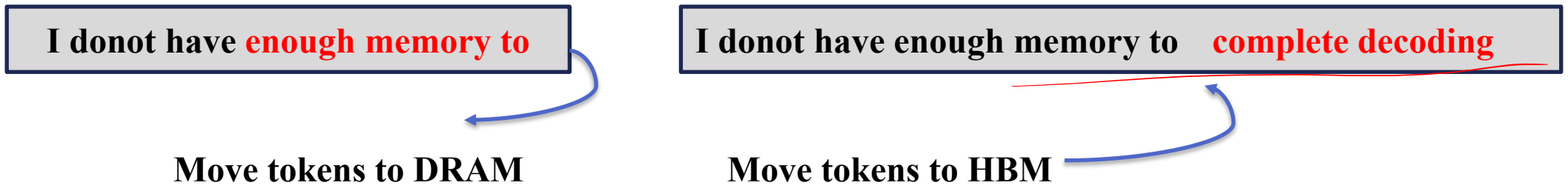
Attention K/V Manager



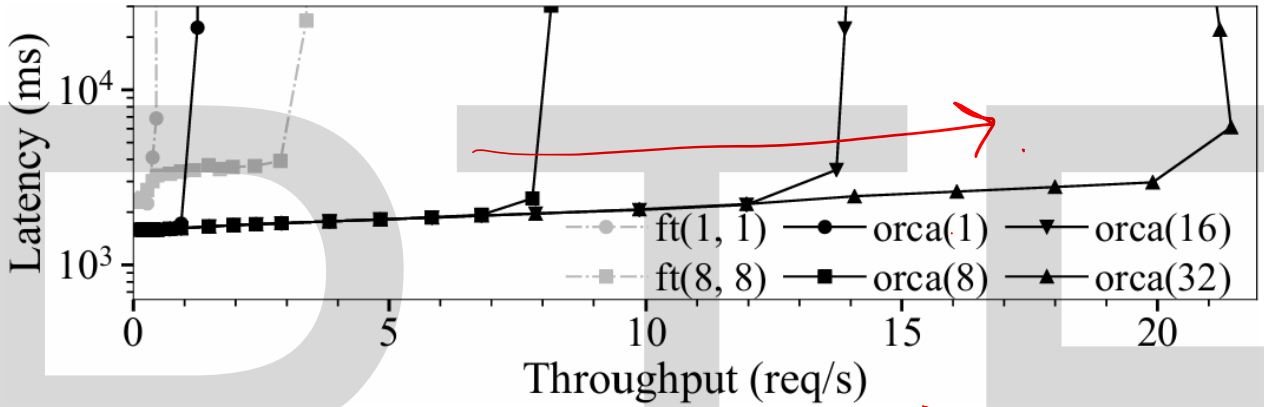
Recall : What can we do better?

- Dynamic memory allocation
- But deadlocks can still happen!
- vLLM : Checkpoint and recompute!

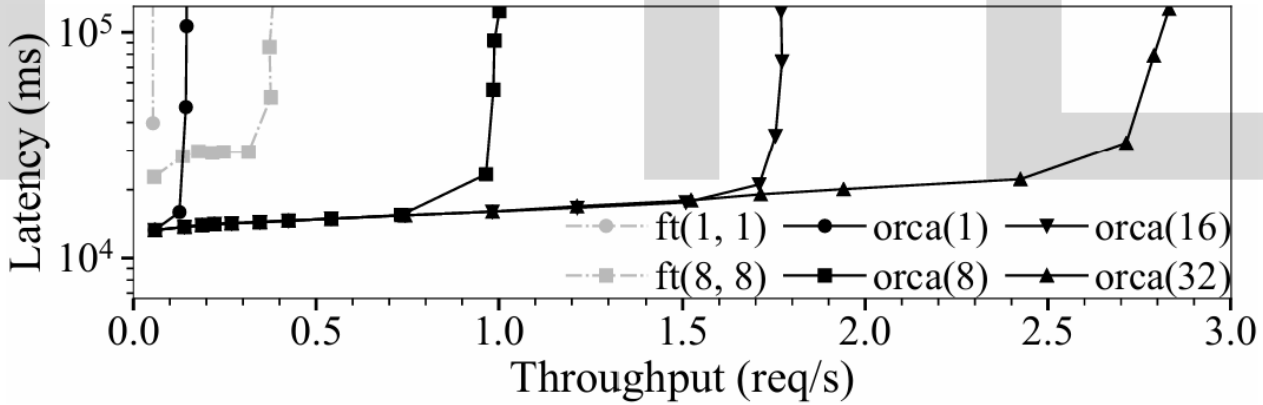
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Results



(a) (# in, # gen) = (32, 32)



(b) (# in, # gen) = (256, 256)

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Takeaway

- ORCA introduced primitives to enable iteration-level scheduling
- But how to meet latency SLOs? [SLO: service-level objective]
- How to schedule requests to improve overall system throughput while meeting their SLO?

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Agenda

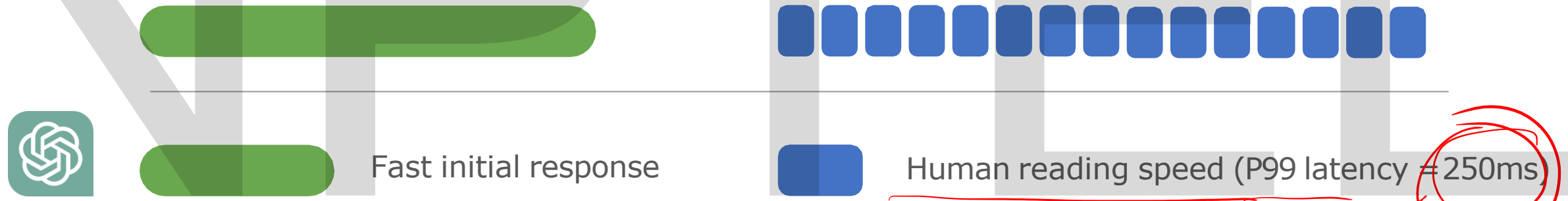
- Orca (contd.)
- **Preliminary to batching**

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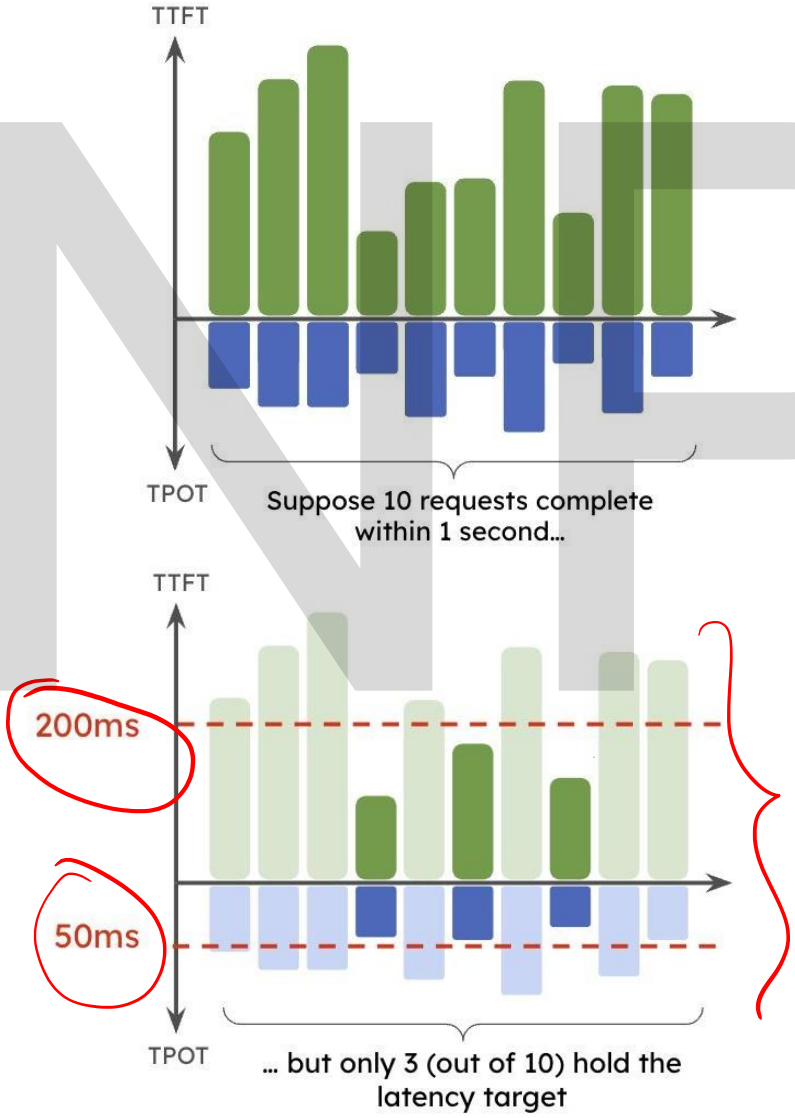
Scheduling with latency-throughput constraints

- **TTFT**
  - Time to first token
  - Initial response time

- **TPOT**
  - Time per output token
  - Average time between two subsequent generated tokens



High Throughput ≠ High Goodput



Throughput = 10 rps
=completed request / time

under SLO criteria

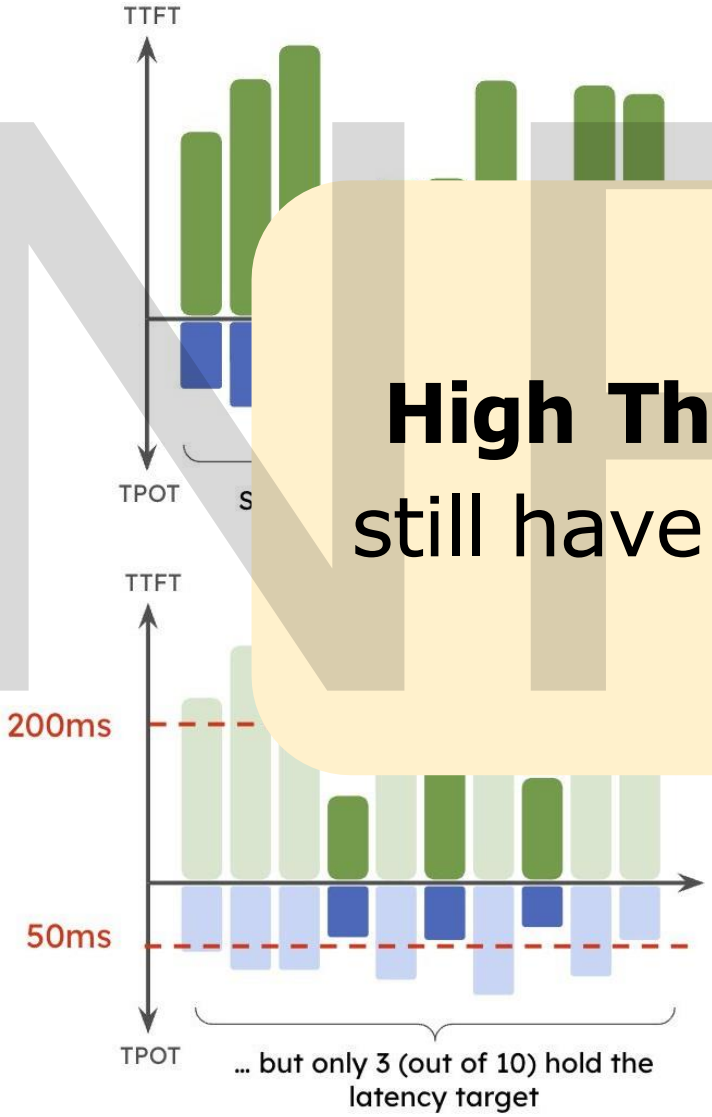
Goodput = 3 rps 🤯
=completed request **within SLO** / time

High Throughput System

...

can have **Low Goodput!**

High Throughput ≠ High Goodput



High

ut

High Throughput can still have **Low Goodput**

~ Poor UX …

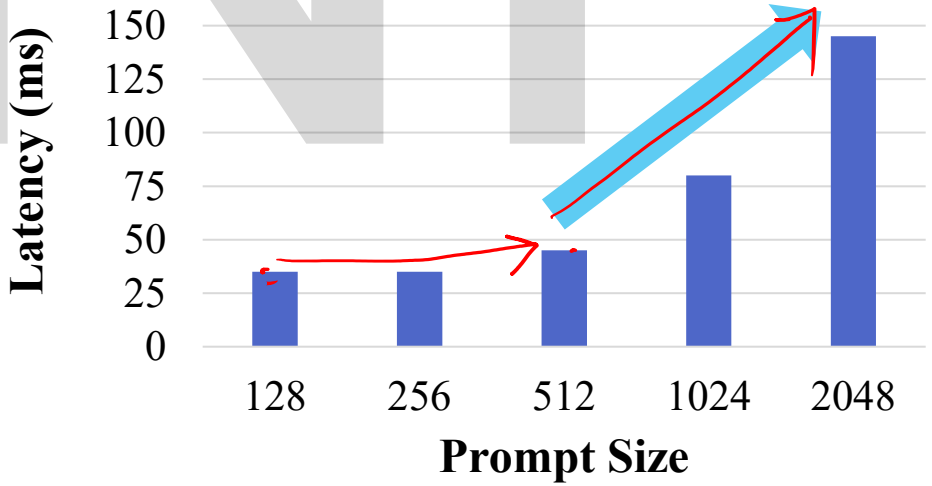
dput!

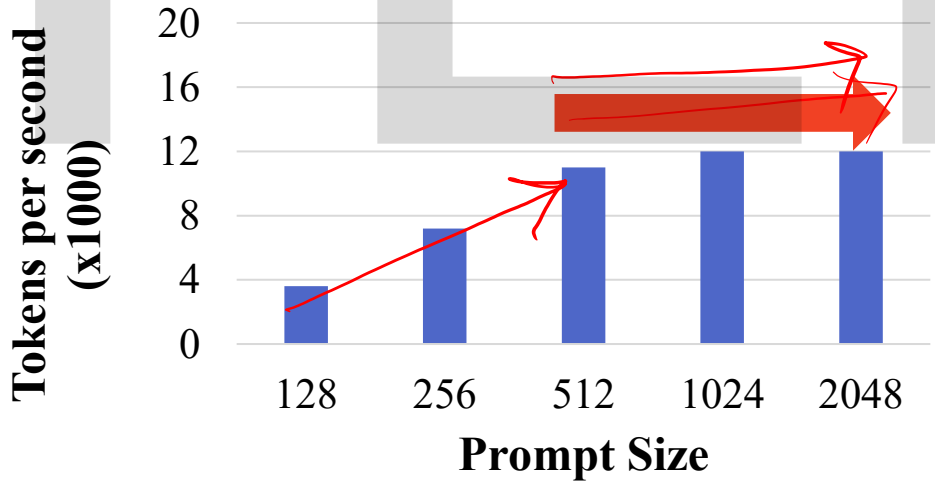
Goodput = 3 rps

=completed request **within SLO** / time

Impact of batching on Prefill phase

- **Excess parallelism**
  - A prompt can contain 1000s of tokens
- **Beyond a certain point, increasing prompt size:**
  - Increases latency, not throughput

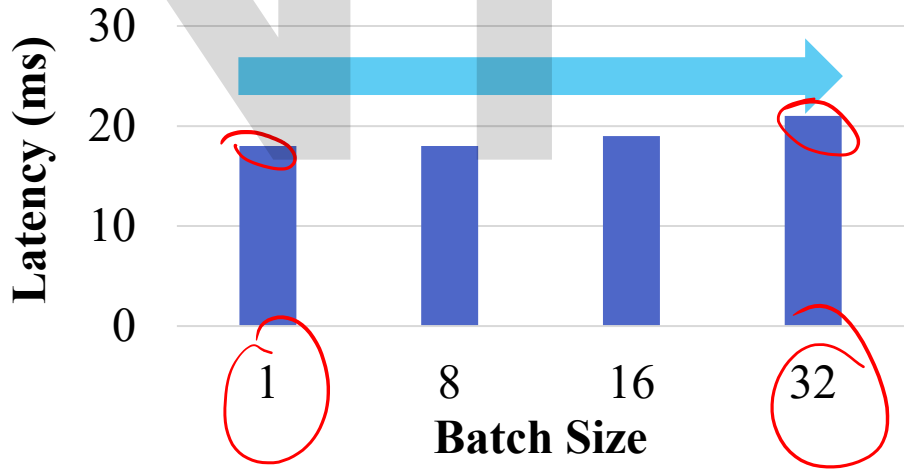


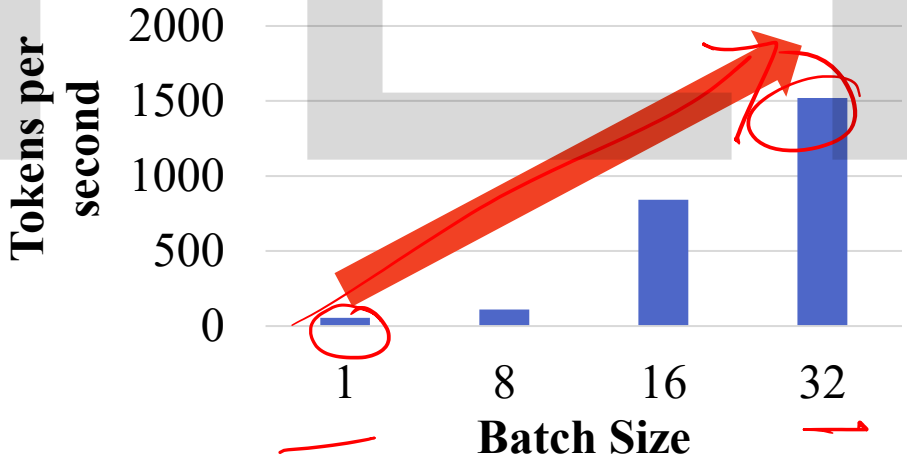


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Impact of batching on Decode phase

- **Lacks parallelism**
  - One token per request
  - – Latency almost entirely dominated by time to read model weights
- **Increasing batch size:**
  - – Increases throughput, not latency





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Scheduling objectives

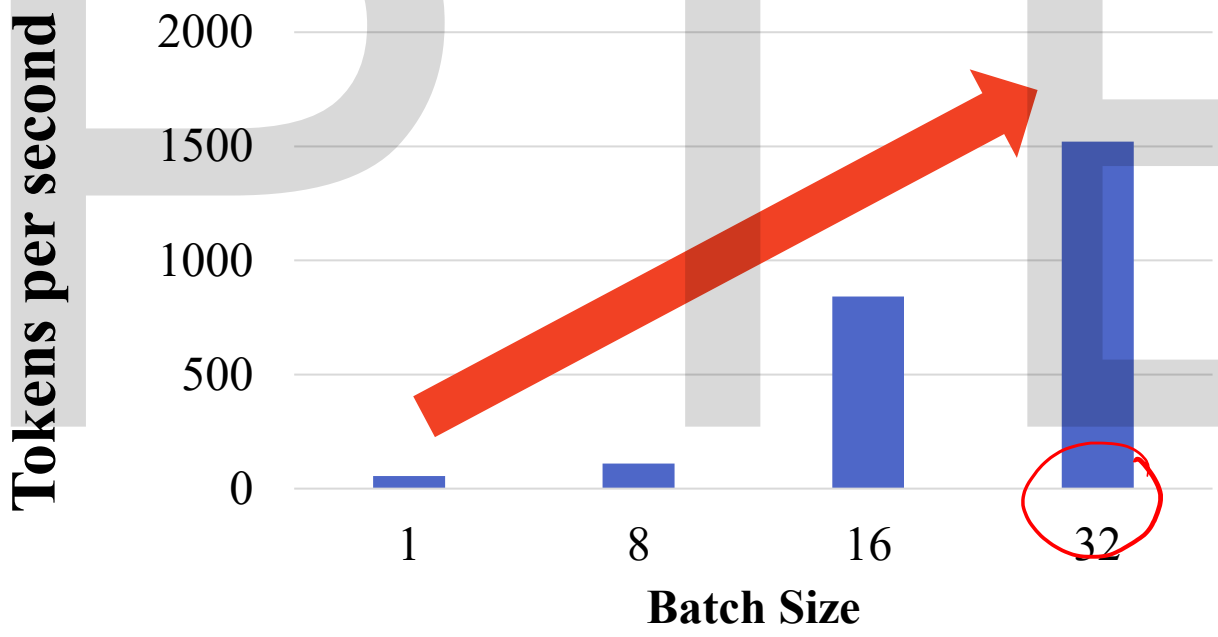
High Throughput + **Low Latency**

- **Time-to-first-token (TTFT)** → Can be somewhat relaxed
  - Scheduling delay + prefill time
- **Time-between-tokens (TBT)** → Constrained
  - Influences request execution time
  - and user experience
- **Serving Capacity**
  - Maximum QPS serviceable under specific latency SLO

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Batching is essential for high throughput

- Because decodes are memory-bound



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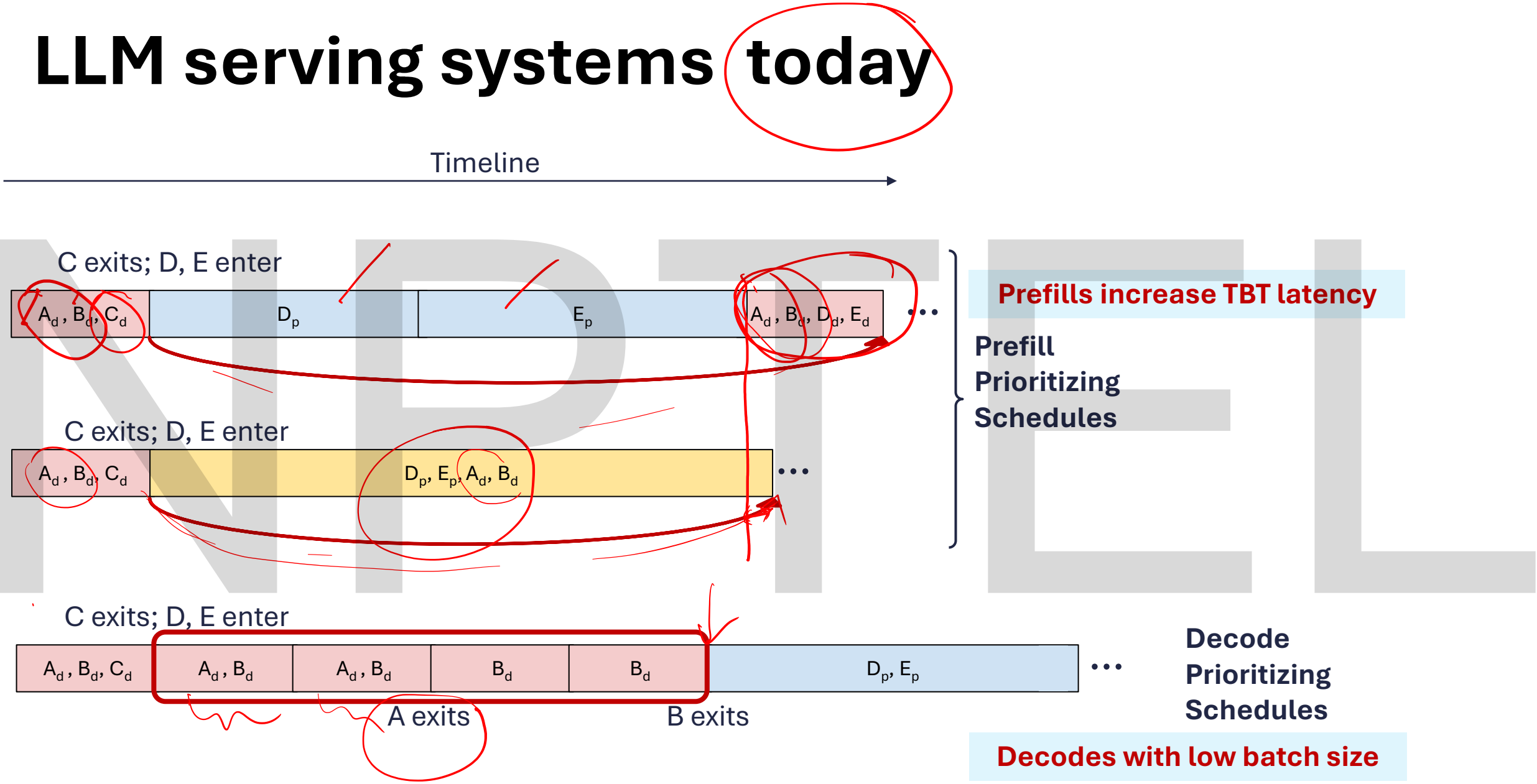
Key scheduling challenge

- **How to increase decode batch size?**
  - A request needs to compute the prefill first
- **Batching involves an interleaving of prefills and decodes**

How to efficiently interleave prefills and decodes?

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LLM serving systems today



Stall free scheduling without compromising on throughput?

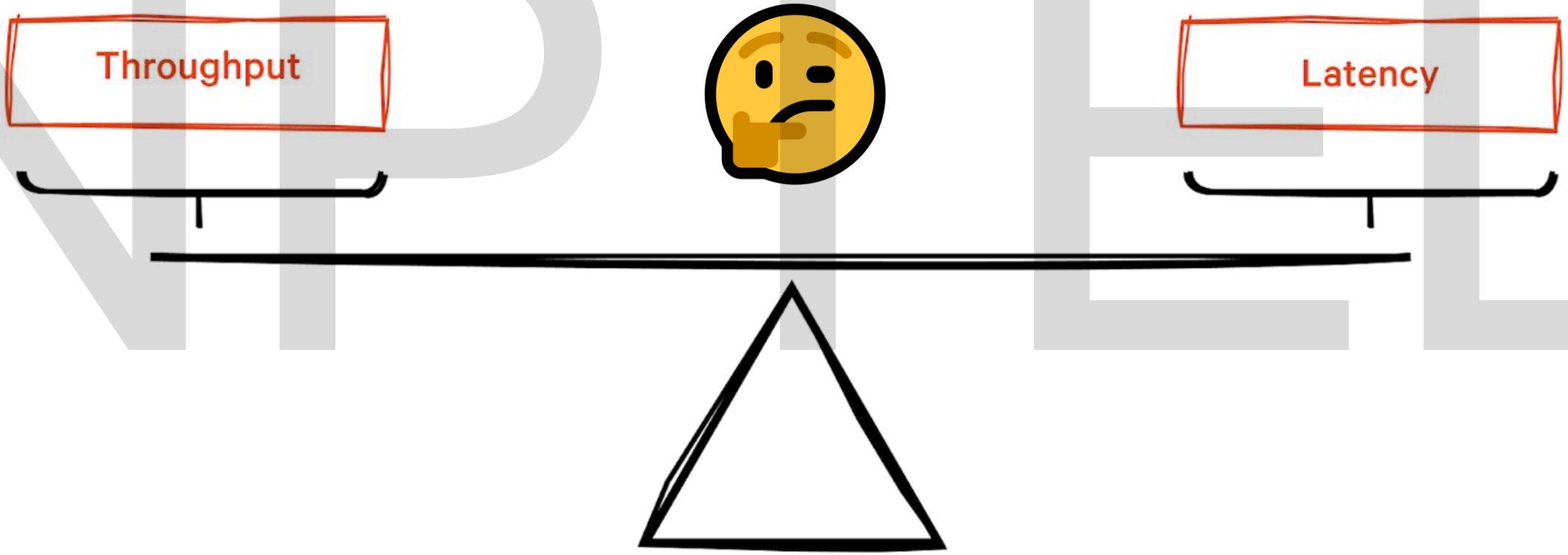
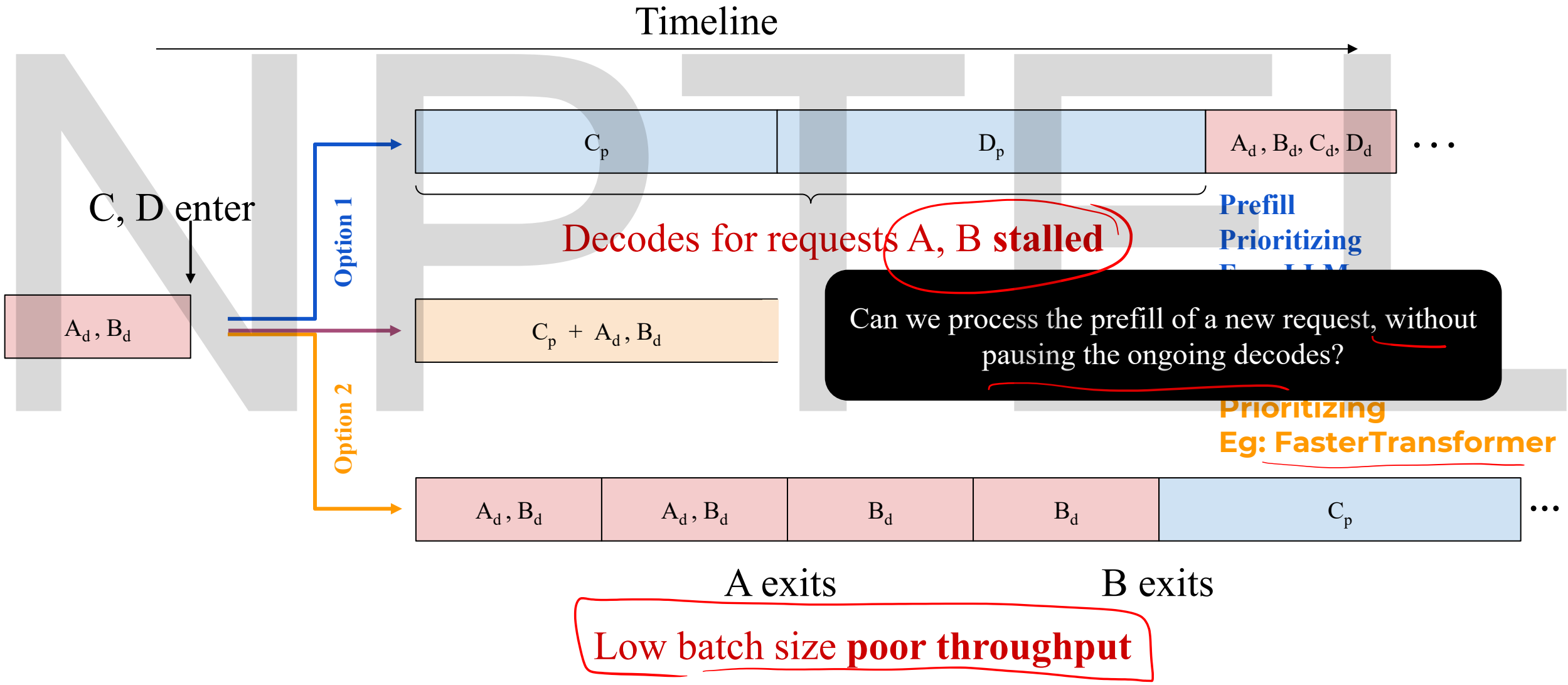


Image credits: Redpanda

How is scheduling done today?





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)